SOURCE CODE REPORT

# Nova Plus Wallet

Self-custody SPA for NovaONE, NRW World, and ALLTRA

| | |
|---|---|
| **Product** | Nova Plus Wallet v1.0.0 |
| **Repository** | `C:\Users\novac\nova-plus-wallet` |
| **Commit** | `0e0b360 — Ship NovaONE and NRW World on live production APIs` |
| **Branch** | `release/v1.0.0` |
| **Author** | novatrusty2-ops |
| **Dated** | 23 August 2026 |
| **Stack** | Vite 8 · React 19 · TypeScript 6 · ethers 6 · Tailwind 3 |
| **Live (Pages)** | https://novatrusty2-ops.github.io/nova-plus-wallet/ |
| **Live (VPS)** | https://novabank.uk/wallet/ (deploy script: `scripts/deploy.ps1`) |
| **Canonical remote** | `github.com/novatrusty2-ops/nova-plus-wallet` |

> This document is a full source-code inventory and architecture report. It describes every application module under `src/`, the production chain/token catalog, custody model, swap and bridge paths, and known production constraints. It is not a dump of every source line.

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## 2. Executive summary

| 119 | 10,352 | 108 | 18 |
|---|---|---|---|
| TypeScript files under src/ | Non-blank source lines | Imported mesh tokens (54 × 2 chains) | Registered chain definitions |

Nova Plus Wallet is a browser self-custody wallet. Keys are generated in the page, encrypted with the user password (ethers JSON keystore / scrypt), and stored only in `localStorage`. The mnemonic is shown once and is not persisted. Signing stays in memory after unlock.

The product surface is send, receive, swap, bridge, token catalog, and settings. NovaONE (chain 22016) and NRW World (33001) are default partner chains. Their catalogs are tokenized onto Nova settlement-meta (m2) liquidity. Local ERC-20 `balanceOf` on those chains is only the wrapped-bridge USDC. ALLTRA and Anaka contract addresses are canonical tokenized references, not copied onto the wrong chain.

Swap on 22016/33001 uses the Nova Bank API (HTTPS host allowlist) with a CoinGecko + peg mid fallback. Public EVM chains use 1inch v6 when `VITE_1INCH_API_KEY` is set. CSS class names still use the historical `signet-*` prefix; the product name is Nova Plus Wallet. Signet is not this product.

## 3. What the product is

### 3.1 Identity

| | |
|---|---|
| **Name** | Nova Plus Wallet (fallback in `src/lib/brand.ts`) |
| **Tagline** | Self-custody wallet for Nova Bank and the Nova mesh |
| **Entry** | `src/main.tsx` → ErrorBoundary → `App.tsx` |
| **Router basename** | `import.meta.env.BASE_URL` so the same build can live at `/wallet/` or `/nova-plus-wallet/` |
| **Dev server** | Vite port 5199 (package.json) / config also lists 5173 — use the npm script |

### 3.2 Dependencies (runtime)

| | |
|---|---|
| **ethers** | ^6.17.0 — RPC, ERC-20, keystore, send, bridge |
| **react / react-dom** | ^19.2.7 |
| **react-router-dom** | ^7.18.1 |
| **recharts** | ^3.9.0 — token chart page |
| **qrcode** | ^1.5.4 — receive address QR |

Dev: Vite 8, TypeScript 6, Tailwind 3, Autoprefixer, PostCSS, @vitejs/plugin-react.

## 4. Architecture

The app is a static SPA. There is no Nova Plus Wallet backend. The browser talks to:

- Public HTTPS RPCs listed on each chain (NovaONE / NRW via novabank.uk plus Railway fallbacks).
- Nova Bank API `/api/v1` (novabank.uk, then Railway) for settlement-meta status and quotes.
- CoinGecko simple price for USD where tokens declare `coingeckoId`.
- 1inch swap API on public EVM, only if an API key is built in.
- Optional chainid.network world-chain list for extra public networks.

#### Layering

| | |
|---|---|
| **UI pages** | `src/pages/` — routes, layout, forms |
| **Components** | `src/components/` — wallet, tokens, swap, bridge, settings, security |
| **Hooks** | `src/hooks/` — balances, enabled chains, FAFO, bridge, transfer |
| **Context** | Wallet (unlock/session) and Toast |
| **Domain lib** | `src/lib/` — chains, keystore, networks, prices, swap, bridge, settlement |

Wallet routes under `/app/*` are gated: no vault → onboarding; locked → `/locked`.

## 5. Source inventory

Counts are non-blank lines from the 23 August 2026 tree.

| Area | Files | Lines | Role |
|---|---|---|---|
| src/lib | 25 | 2,727 | Chains, keystore, prices, token lists, settlement, networks |
| src/pages | 14 | 1,833 | Landing, dashboard, send/receive/swap, tokens, assets, bridge |
| src/pages/settings | 14 | 961 | Accounts, security, backup, network, display, about |
| src/hooks | 8 | 797 | Live reads and polling |
| src/lib/bridge | 6 | 641 | Hub/spoke/vault execute + history |
| src/components/tokens | 5 | 472 | Catalog, search, add, hide |
| src/components/bridge | 3 | 394 | Bridge UI |
| src/lib/swap | 3 | 334 | Settlement-meta, 1inch, global-swap URL |
| src/context | 2 | 309 | Wallet session + toasts |
| src/components/assets | 2 | 293 | Asset list and detail |
| src/components/wallet | 4 | 263 | Connect, account, network switchers |
| src/components/common | 11 | 320 | Shell, modal, empty/error, QR-adjacent UI |
| Other components | — | ~650 | Layout, security, settings, FAFO, banks, swap widget |
| src root | 4 | 156 | App, main, ErrorBoundary, CSS entry |

### 5.1 Largest modules

| Lines | File |
|---|---|
| 452 | `src/pages/SwapPage.tsx` |
| 401 | `src/pages/settings/SettingsNetworkPage.tsx` |
| 370 | `src/pages/SendPage.tsx` |
| 287 | `src/context/WalletContext.tsx` |
| 280 | `src/hooks/useTokenBalances.ts` |
| 267 | `src/lib/networks.ts` |
| 255 | `src/lib/bridge/execute.ts` |
| 235 | `src/components/assets/AssetDetail.tsx` |
| 224 | `src/lib/chains.ts` |
| 221 | `src/lib/keystore.ts` |
| 215 | `src/lib/transfer.ts` |
| 206 | `src/lib/prices.ts` |
| 194 | `src/lib/swap/metaSettlement.ts` |

## 6. Routes and screens

Defined in `src/App.tsx` and `src/lib/routes.ts`.

| Path | Gate | Screen |
|---|---|---|
| `/` | public | LandingPage |
| `/onboarding` | public | Create / import wallet |
| `/locked` | vault exists | Password unlock |
| `/app` | unlocked | Dashboard — portfolio + token rows |
| `/app/send` | unlocked | Native / ERC-20 send |
| `/app/receive` | unlocked | Address + QR |
| `/app/swap` | unlocked | Settlement-meta or 1inch |
| `/app/tokens` | unlocked | Balances + mesh catalog + search/add |
| `/app/tokens/chart/:id` | unlocked | CoinGecko chart (recharts) |
| `/app/assets` | unlocked | Asset list |
| `/app/assets/:chainId/:symbol` | unlocked | Asset detail |
| `/app/bridge` | unlocked | Hub / spoke bridge UI |
| `/app/banks` | unlocked | Bank directory placeholders |
| `/app/more` | unlocked | More links |
| `/app/settings/*` | unlocked | Accounts, general, security, backup, network, notifications, display, connections, advanced, FAFO, about |

Legacy unprefixed `/tokens`, `/assets`, `/bridge`, `/settings` redirect into `/app`.

## 7. Chain registry

Source: `src/lib/chains.ts`. Values marked `NEEDS_CONFIRM` fail loud instead of inventing data. NovaONE is **partner / default** (not private-banking), so it is visible without the institutional gate.

| ID | Name | Tier | Default | Notes |
|---|---|---|---|---|
| 22016 | NovaONE | partner | yes | Settlement-meta; RPC novabank.uk + Railway |
| 33001 | NRW World | partner | yes | Same liquidity rail; local USDC wrap |
| 138 | Defi Oracle Meta | partner | yes | 17 verified ERC-20s |
| 651940 | ALLTRA | partner | yes | Canonical tokenized ERC-20 home for many mesh refs |
| 9001 | Nova Production | partner | no | Internal mesh |
| 11011 | AnakaChain | public | no | |
| 11012–11015 | Anaka layers | private-banking | no | Hidden until institutional gate |
| 1, 56, 42161, 8453, 137, 10, 43114, 100 | Public EVM | world | no | 1inch-capable when key present |

`src/lib/networks.ts` persists enabled chain IDs in `signet.networks.v1`. A one-time migration key `signet.networks.settlement-v1` inserts 22016 and 33001 for wallets that saved an older enabled-set. Private-banking chains are filtered unless the session gate is unlocked. RPC validation uses ethers `JsonRpcProvider` and checks `eth_chainId`.

## 8. Token catalog (NovaONE / NRW)

Generated catalog: `src/lib/nova-mesh-tokens.ts` (from `public/nova-chains.json` via `scripts/gen-mesh-tokens.cjs`). Runtime does **not** fetch that JSON; the TypeScript module is the live catalog.

| | NovaONE 22016 | NRW World 33001 |
|---|---|---|
| Listed assets | 54 | 54 |
| Native | 1 (NOVA) | 1 (NRW) |
| Local ERC-20 wrap | 1 USDC | 1 USDC |
| Meta / tokenized | 52 | 52 |
| Canonical tokenizedAddress | 45 | 45 |
| Swap router | `0xEd04ee8307C0656207af5afe3926Ae2380052940` | |
| Wrapped bridge (USDC) | `0x227a6f3EEF2df576bc127dF0326D1F4B6f0ce9Cb` | `0xf2811B06807acF321FF9056dF668a621772999d8` |

Canonical refs: ALLTRA 651940 ERC-20s, and Anaka wrap 11013 for USD, EUR, GBP, AUD, CHF, JPY, USDT, ETH, BTC. Those addresses must not be used as `balanceOf` targets on 22016/33001. `src/lib/tokenlist.ts` skips `standard === "meta"` when building local ERC-20 lists. UI: `MeshTokenCatalog.tsx` on the Tokens page.

## 9. Custody and security

Source: `src/lib/keystore.ts`.

- Create: `Wallet.createRandom()` (BIP-39).
- Import: 12/24-word phrase or 32-byte hex private key.
- Persist: `wallet.encrypt(password)` → `signet.keystore.v1`. Password minimum 8 characters.
- Metadata: address, createdAt, accountType in `signet.meta.v1`.
- Unlock: `Wallet.fromEncryptedJson`; signer held in React context only.
- Auto-lock: `AutoLockMonitor` clears the in-memory signer after idle.

> **Institutional gate.** `src/lib/institutionalGate.ts` compares a passphrase to SHA-256(`VITE_INSTITUTIONAL_GATE`) in the client. The file documents that this is not server-side authorization. Private-banking Anaka layers stay hidden until the session flag is set.

Nova API fetches in `src/lib/nova-api.ts` allow only HTTPS hosts: novabank.uk, www, Railway production, plus a build-time `VITE_NOVA_API_URL` host. Relative `/api/v1` is allowed only on those same-origin hosts. This is the SSRF control for settlement quotes.

## 10. Balances, prices, and lists

`useTokenBalances` walks enabled reachable chains. For each chain it tries RPC URLs in order until one answers. It reads native balance and ERC-20 `balanceOf` only for tokens with a local address and `standard === "erc20"` (plus curated public lists / user-added tokens). Timeout 8s per RPC ping.

USD: token `usd` peg → CoinGecko id → oracle map in `src/lib/prices.ts` (60s cache). Public chain token lists: Uniswap, PancakeSwap, and other remote lists in `tokenlist.ts`, cached 24h. User tokens: `usertokens.ts`. Hidden tokens: prefs.

## 11. Swap: settlement-meta and 1inch

### 11.1 NovaONE / NRW (settlement-meta)

If the selected swap chain is 22016 or 33001, `SwapPage` renders `MetaSettlementSwap`.

- Status: GET chain status, GET `/swap/status`, GET routers, NRW hub when on 33001. Probe API bases until `/chains/nova-one` returns JSON.
- Pairs: `src/lib/nova-one-markets.ts` builds Binance-style symbols (e.g. USDC+NOVA → USDCNOVA). SELL if from is base; BUY if from is quote.
- Quote: POST `/chains/nova-one/markets/quote`. Parse `price`, `cummulativeQuoteQty` / `amountOut`.
- Fallback: USD cross via pegs + CoinGecko, 30 bps haircut, labelled settlement-mid. Used when quote POSTs 503 (observed on Railway hub/quote in August 2026).
- Execute: this wallet does **not** submit settlement ledger orders. Copy states ledger execute stays on Nova Bank API.

### 11.2 Public EVM (1inch)

`src/lib/swap/oneinch.ts` — v6 quote + swap tx. Mesh chain IDs including 22016 and 33001 are in the unsupported set so 1inch is never faked there. `globalSwap.ts` can point at an in-app global swap URL for mesh chains; NovaONE/NRW UI prefers settlement-meta.

CORS: GitHub Pages calling Railway may be blocked. Production on novabank.uk should use same-origin `/api/v1`.

## 12. Bridge

Module `src/lib/bridge/`: constants, ABIs, route table, token map, history in localStorage, `execute.ts` (approve + lock/burn via ethers). UI: Bridge page, panel, review modal, history. Hub/spoke/vault addresses come from the registry — CONFIRM markers must not be guessed. This is separate from settlement-meta tokenization.

## 13. Build, brand, and deploy

| | |
|---|---|
| **Brand tokens** | `branding/brand-tokens.json` → `src/lib/brand.generated.ts` via `scripts/sync-branding.mjs` |
| **Vite base** | `VITE_BASE_PATH` in `vite.config.ts` |
| **VPS deploy** | `scripts/deploy.ps1` — build `/wallet/`, scp to `ubuntu@51.75.64.28:/var/www/nova-plus-wallet` |
| **GitHub Pages** | dist-only `gh-pages` branch, base `/nova-plus-wallet/`. Do not nest Signet. |
| **Env example** | `.env.example` — institutional gate hash, 1inch key, `VITE_NOVA_API_URL` |

On 23 August 2026 the VPS scp from this Windows OpenSSH client failed (host KEX `sntrup761x25519`). Pages was updated. novabank.uk/wallet/ may still serve an older JS hash until deploy.ps1 succeeds from WSL or a newer ssh.

## 14. Findings and production notes

| Item | Severity | Detail |
|---|---|---|
| Institutional gate is client-only | High (if used as real ACL) | Hash is in the frontend bundle. Hides UI only. |
| Settlement quotes may 503 | Medium | Wallet falls back to peg/CoinGecko mid. Not a live order book when fallback is used. |
| No in-wallet settlement execute | Info | Quotes only on meta rails. Intentional for self-custody SPA. |
| localStorage keystore | Info | Standard web-wallet model. XSS would be catastrophic; no extension sandbox. |
| signet-* CSS and storage keys | Low | Historical names; product is Nova Plus Wallet. |
| Wallet JSON is not loaded from nova-chains.json at runtime | Info | Regenerate `nova-mesh-tokens.ts` when the mesh JSON changes. |
| GitHub Pages CORS | Medium | Prefer novabank.uk/wallet/ for live API. |

## 15. Appendix — every src TypeScript file

Non-blank line counts. 119 files, 10,352 lines.

| Lines | Path |
|---|---|
| 122 | src/App.tsx |
| 12 | src/main.tsx |
| 35 | src/components/ErrorBoundary.tsx |
| 235 | src/components/assets/AssetDetail.tsx |
| 58 | src/components/assets/AssetList.tsx |
| 20 | src/components/banks/BanksButton.tsx |
| 101 | src/components/bridge/BridgeHistory.tsx |
| 232 | src/components/bridge/BridgePanel.tsx |
| 61 | src/components/bridge/BridgeReviewModal.tsx |
| 320 | src/components/common/* (11 files) |
| 79 | src/components/dashboard/PortfolioAnchor.tsx |
| 64 | src/components/fafo/* |
| 139 | src/components/layout/* |
| 118 | src/components/security/* |
| 250 | src/components/settings/* |
| 124 | src/components/swap/MetaSettlementSwap.tsx |
| 472 | src/components/tokens/* |
| 263 | src/components/wallet/* |
| 22 | src/context/ToastContext.tsx |
| 287 | src/context/WalletContext.tsx |
| 192 | src/hooks/useBridge.ts |
| 66 | src/hooks/useChainData.ts |
| 18 | src/hooks/useDisplaySettings.ts |
| 102 | src/hooks/useEnabledChains.ts |
| 49 | src/hooks/useFAFO.ts |
| 280 | src/hooks/useTokenBalances.ts |
| 66 | src/hooks/useTokenTransfer.ts |
| 24 | src/hooks/useVisibleTokenRows.ts |
| 135 | src/lib/accounts.ts |
| 44 | src/lib/activeChain.ts |
| 53 | src/lib/activity.ts |
| 224 | src/lib/chains.ts |
| 221 | src/lib/keystore.ts |

| | |
|---|---|
| 267 | src/lib/networks.ts |
| 104 | src/lib/nova-api.ts |
| 122 | src/lib/nova-mesh-tokens.ts |
| 65 | src/lib/nova-one-markets.ts |
| 206 | src/lib/prices.ts |
| 76 | src/lib/settlement-meta.ts |
| 194 | src/lib/swap/metaSettlement.ts |
| 110 | src/lib/swap/oneinch.ts |
| 30 | src/lib/swap/globalSwap.ts |
| 231 | src/lib/tokenlist.ts |
| 215 | src/lib/transfer.ts |
| 641 | src/lib/bridge/* (6 files) |
| 1,833 | src/pages/* (14 files, excl. settings) |
| 961 | src/pages/settings/* (14 files) |
| 23 | src/types/index.ts |

Supporting non-src: public/nova-chains.json, public/manifest.json, vite.config.ts, scripts/deploy.ps1, scripts/gen-mesh-tokens.cjs, scripts/sync-branding.mjs, branding/.

End of report · Generated 23 August 2026 from commit 0e0b360 on release/v1.0.0 · Nova Plus Wallet